GlucoTwin — Digital Twin for Type 2 Diabetes

System Architecture Diagram • Real-time Predictive Healthcare Platform

LAYER 1 • DATA SOURCES

Wearable IoT Sensors

- CGM (Continuous Glucose Monitor)
- Heart Rate
- Steps
- Sleep

Static EHR

- Synthea
- Age, BMI, HbA1c

• LAYER 2 • DATA FUSION LAYER

FastAPI Backend

- Data Validation & Normalization
- Time-Series Alignment & Feature Engineering
- Secure API Gateway (REST, JSON)

• LAYER 3 • AI PREDICTION ENGINE

LSTM Model

- Predicting glucose spike in 2 hours
- Risk Score: Low • Medium • High
- Temporal Sequence Modeling • Real-time Inference

• LAYER 4 • DOCTOR DASHBOARD

Virtual Patient
Digital Twin State

Live Graphs
Glucose & Vitals Trends

Alert System
Threshold Alerts & Notifications

What-If Simulator
Intervention Scenario Testing

Flow: Data Ingestion → Processing → AI Inference → Clinical Dashboard

GlucoTwin v1.0 • HIPAA-Compliant • Secure Data Flow • Cloud Deployable